

The code for \$a_2\$ is:

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Yes, the point $\{0.4, 0.6\}$ is at most 0.45 away from a minimiser in the infinity norm.

The number a_2 is the limit of the computable sequence defined by the code. The infinite sum in $f(n)$ tends to

$$f(\infty) = \frac{1}{4} + 72^{-B} - \frac{72^{-3B}}{1 - 72^{-B}},$$

where $B = 10^{24}$. Hence

$$a_2 = 2f(\infty) + \frac{1}{2} = 1 + 2 \left(72^{-B} - \frac{72^{-3B}}{1 - 72^{-B}} \right) > 1,$$

and the positive term is astronomically small (so a_2 is only fractionally larger than 1).

The optimisation problem is: minimise $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$. Since $a_2 > 1$, the unique minimiser is $x_1^* = 0$, $x_2^* = 1/a_2 \approx 1$.

The infinity-norm distance from $(0.4, 0.6)$ to $(0, 1/a_2)$ is

$$\max(|0.4 - 0|, |0.6 - 1/a_2|).$$

Because $1/a_2 < 1$ but extremely close to 1, $|0.6 - 1/a_2| = 1/a_2 - 0.6 < 0.4$. Thus the distance is exactly 0.4, which is less than 0.45.